9. Which of the following function can be used to detect whether a car exceed the speed limit on a highway?

A) Parity generator

B) Comparator

C) Encoder

D) Multiplexer

10. What is the output of the priority decoder in Figure 2 if the input is HIGH at 8, 6, 3 and 1 and all other input are LOW?

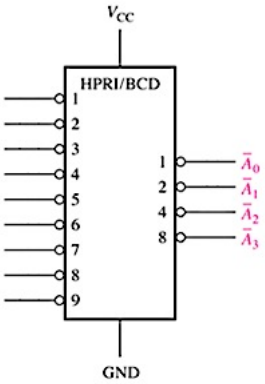


Figure 2

A) 1000_2

B) 0110_2

C) 1111_2 to indicate an error

D) 0000_2 to indicate an error

Questions 11 and 12 referring to circuit in Figure 3.

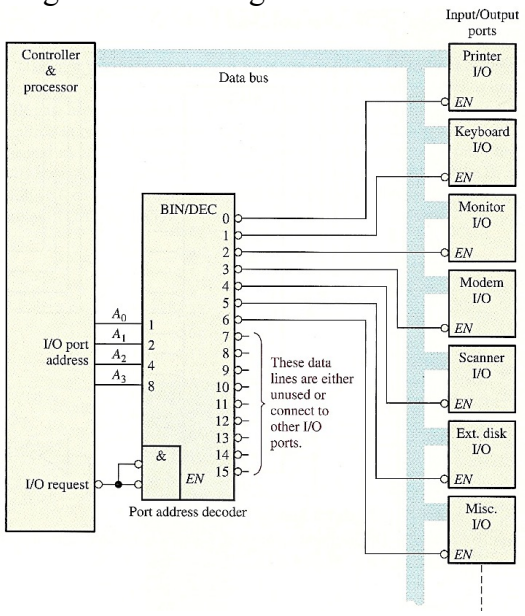


Figure 3

11. What is the function of IC labeled as BIN/DEC in Figure 3?

A) Encoder

B) DeMultiplexer

C) Code converter

D) Decoder

12. What is the value of $A_3A_2A_1A_0$ to select the Scanner if I/O request equals to 0 in Figure 3?

A) 0100_2

B) 0110_2

C) 1001_2

D) 0011_2

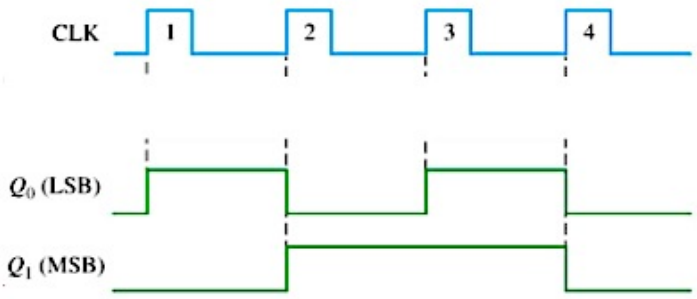
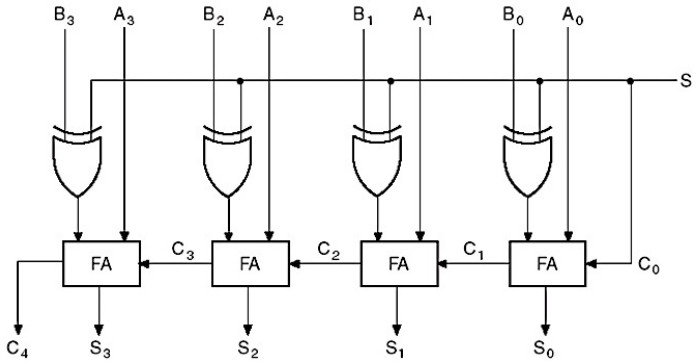


Figure 4



26. Which function can be used to route a signal from one place to the other?

    i. Multiplexer.

    ii. DeMultiplexer.

    iii. Decoder.

    iv. Code converter.

    A) i and ii.

    B) i, ii and iii.

    C) iii and iv.

    D) All of the above.

(b) i) Design and draw a 3-bit unsigned parallel adder using full adder. [3 marks]

ii) On the circuit shows the value at the input and output while adding two numbers which the value are A = 101_2 and B = 110_2. [2 marks]